



Specifications of Welded Steel

ASTM A795 Standard Specification for Black and Hot-Dipped Galvanized Welded steel pipe for fire Protection Use

Schedule	Nominal Size	Outside Diameter	Wall Thickness	Mass of Black Tube	Test Pressure
	(inch)	(mm)	(mm)	Plain End (kg/m)	Grade B (PSI)
Schedule 10	3/4"	26.7	2.11	1.28	700
	1"	33.4	2.77	2.09	700
	1 1/4"	42.2	2.77	2.69	1000
	1 1/2"	48.3	2.77	3.11	1000
	2"	60.3	2.77	3.93	1000
	2 1/2"	73.0	3.05	5.26	1000
	3"	88.9	3.05	6.46	1000
	4"	114.3	3.05	8.37	1200
	5"	141.3	3.40	11.58	1200
	6"	168.3	3.40	13.85	1000
	8"	219.1	4.78	25.26	800
	10"	273	4.78	31.62	700

ASTM A53 Standard Specification for Pipe, Steel, Black and Hot-Dipped, Welded.

Schedule	Nominal Size	Outside Diameter	Wall Thickness	Mass of Black Tube	Test Pressure
	(inch)	(mm)	(mm)	Plain End (kg/m)	Grade B (PSI)
Schedule 40	1/2"	21.3	2.77	1.27	700
	3/4"	26.7	2.87	1.69	700
	1"	33.4	3.38	2.5	700
	1 1/4"	42.2	3.56	3.39	1300
	1 1/2"	48.3	3.68	4.05	1300
	2"	60.3	3.91	5.44	2500
	2 1/2"	73	5.16	8.63	2500
	3"	88.9	5.49	11.29	2500
	4"	114.3	6.02	16.07	2210
	5"	141.3	6.55	21.77	1950
	6"	168.3	7.11	28.26	1780
	8"	219.1	8.18	42.55	1570
	10"	273	9.27	60.29	1430
	12"	323.8	10.31	79.7	1340

Note: Wall thickness tolerance is -12.5%

Length tolerance is +20mm/0

Welded Steel Pipe



Raw Material: Hot Rolled Coil

Forming Method: F-F-X

Welding Method: High-frequency Resistance Welding

Size: 1/2" ---8"

Temperature: -30°C--80°C

Normal Pressure: 0.1 Mpa--8Mpa

Connection: Grooved, Threaded, Beveled, Plain

Application: Water System, Sprinkler System, Gas system, Oil and HAVC System

Product Advantages:

Advanced F-F-X forming technics, enhance the controllability.

Thermatool solid state high frequency welding Machine, make sure the welding stability.

On line Heat Treatment, make sure the seam hardness

100% on line Eddy Current Testing and Hydrostatic Testing, make sure the welding quality.

Painted Steel Pipe



Coating Material: Anti-rust Paint, Primer Paint, Epoxy Paint

Coating Color: Red, Black, Gray, Varnish

Coating Thickness: 30um

Coating Method: Painted Outside

Size: 1" ---8"

Temperature: -30°C--80°C

Normal Pressure: 0.1 Mpa--8Mpa

Connection: Grooved, Threaded, Plain

Application: Sprinkler System, Gas system

Product Advantages:

Paint is high quality and environmental, meet the requirement of usage

Coating is uniformity and smooth.

Coating with good adhesion and hardness

Coating with good anticorrosion, prolong the pipe life span.

Galvanized Steel Pipe



Galv.: Hot-dip Galvanizing

Size: 1/2" ---8"

Temperature: -30°C--80°C

Normal Pressure: 0.1 Mpa--8Mpa

Connection: Grooved, Threaded, Beveled, Plain

Application: Water System, Sprinkler System, Gas system, Oil and HAVC System

Product Advantages:

Galv. coating is uniformity, clean and bright.

Using 0# zinc ingot, with good anti-corrosion ability.

Expoxy Steel Pipe



Coating Material: Epoxy Resin Powder

Coating Color: Red, Blue

Coating Thickness: 40um-300um

Size: 1/2" ---8"

Temperature: -30°C--80°C

Normal Pressure: 0.1 Mpa--5Mpa

Connection: Grooved, Threaded, Plain

Application: Water System, HAVC System

Product Advantages:

Epoxy powder is high-quality and environmental

Coating with good anti-corrosion

Coating with good hygienism

Small resistance

Steel Pipe Nipple with American Standard

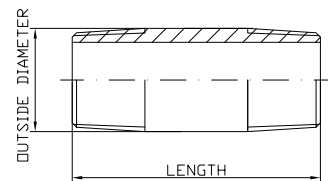


Material : ASTM A53/A106/A333

Dimensions : ASTM A733

Threads : ASME B1.20.1

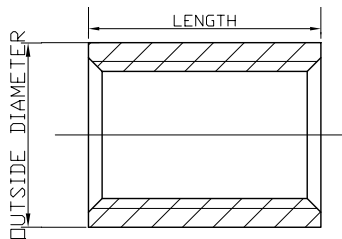
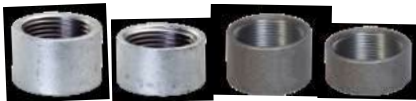
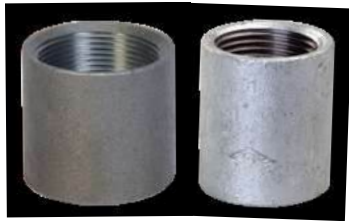
Schedule : 40/80/160/XXS



Carbon Steel Pipe Nipple Sch.40/80/160/XXS, Welded/Seamless																		
Pipe Size	Pipe O.D.	Length Close	Pipe Nipple Lengths (In)															
In	In	In																
1/8	0.405	3/4	1 1/2	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
1/4	0.540	7/8	1 1/2	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
3/8	0.675	1	1 1/2	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
1/2	0.840	1 1/8	1 1/2	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
3/4	1.050	1 3/8	1 1/2	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
1	1.315	1 1/2	1 1/2	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
1-1/4	1.660	1 5/8	X	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
1-1/2	1.900	1 3/4	X	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
2	2.375	2	X	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
2-1/2	2.875	2 1/2	X	X	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
3	3.500	2 5/8	X	X	X	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
4	4.500	2 7/8	X	X	X	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
5	5.563	3	X	X	X	3	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
6	6.625	3 1/8	X	X	X	X	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12
8	8.625	3 1/2	X	X	X	X	3 1/2	4	4 1/2	5	5 1/2	6	7	8	9	10	11	12

Noted: Other lengths and size available. Grooved end, SQ end and Bevel end available. Contact us for details.

Steel Merchant Coupling with American Standard



Material : ASTM A865

Dimensions : ASTM A865

Threads : ASME B1.20.1

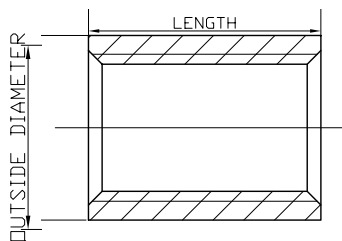
Schedule : Standard

Size		In	1/8	1/4	3/8	1/2	3/4	1	1 1/4	1 1/2
Full	Pipe O.D.	In	0.563	0.719	0.875	1.063	1.313	1.576	1.900	2.200
	Length	In	3/4	1-1/8	1-1/8	1-1/2	1-9/16	1-15/16	2	2
Half	Pipe O.D.	In	0.563	0.719	0.875	1.063	1.313	1.576	1.900	2.200
	Length	In	0.335	0.531	0.531	0.709	0.748	0.925	0.965	0.965

Size		In	2	2 1/2	3	3 1/2	4	5	6
Full	Pipe O.D.	In	2.75	3.250	4.000	4.625	5.000	6.296	7.390
	Length	In	2-1/16	3-1/16	3-3/16	3-5/16	3-7/16	3-11/16	3-15/16
Half	Pipe O.D.	In	2.75	3.250	4.000	4.625	5.000	6.296	7.390
	Length	In	1.051	1.496	1.555	1.634	1.673	1.811	1.949

Noted: Other lengths and size available, contact us for details

Steel Merchant Coupling with American Standard



Material : ASTM A865

Dimensions : ASTM A865

Threads : ASME B1.20.1

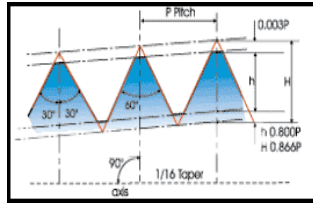
Schedule : Extra-Strong

Size		In	1/8	1/4	3/8	1/2	3/4	1	1 1/4	1 1/2
Standard	Pipe O.D.	In	0.563	0.719	0.875	1.063	1.313	1.576	2.054	2.2
	Length	In	1-1/18	1-5/8	1-5/8	2-1/8	2-1/8	2-5/8	2-3/4	2-3/4

Size		In	2	2 1/2	3	3 1/2	4	5	6	
Standard	Pipe O.D.	In	2.875	3.375	4.0	4.625	5.2	6.296	7.39	
	Length	In	2-7/8	4-1/8	4-1/4	4-3/8	4-1/2	4-5/8	4-7/8	

Noted: Other lengths and size available, contact us for details

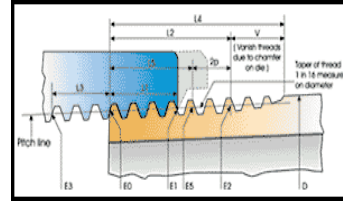
Basic Dimention for American Standard Pipe Threads



Taper thread

$$H = 0.866025 \quad p = 0.866025/n$$

$$h = 0.800 \quad p = 0.800/n$$



Taper thread

$$L2 = (0.8D + 6.8) \frac{1}{n} = (0.80D + 6.8)p$$

Nominal Pipe Size(inchs)	O.D.of Pipe (D)	Threads /in (N)	Pitch of Thread (P)	Pitch Dia. at Beginning of External Thread(E0)	Handtight Engagement			Effective Thread.Extenal			Depth of Thread h
					Length² (L1)		Diam³. (E1)	Length⁴ (L2)		Diam. (E2)	
					Inch	Thread		Inch	Thread		
1/8	0.405	27.0	0.03704	0.36351	0.1615	4.36	0.37360	0.2639	7.12	0.38000	0.029
1/4	0.540	18.0	0.05556	0.47739	0.2278	4.10	0.49163	0.4018	7.23	0.50250	0.044
3/8	0.675	18.0	0.05556	0.61201	0.2400	4.32	0.62701	0.4078	7.34	0.63750	0.044
1/2	0.840	14.0	0.07143	0.75843	0.3200	4.48	0.77843	0.5337	7.47	0.79179	0.057
3/4	1.050	14.0	0.07143	0.96768	0.3390	4.75	0.98887	0.5457	7.64	1.00179	0.057
1	1.315	11.5	0.08696	1.21363	0.4000	4.60	1.23863	0.6826	7.85	1.25630	0.069
1 1/4	1.660	11.5	0.08696	1.55713	0.4200	4.83	1.58338	0.7068	8.13	1.60130	0.069
1 1/2	1.900	11.5	0.08696	1.79609	0.4200	4.83	1.82234	0.7235	8.32	1.84130	0.069
2	2.375	11.5	0.08696	2.26902	0.4360	5.01	2.29627	0.7565	8.70	2.31630	0.069
2 1/2	2.875	8.0	0.12500	2.71953	0.6820	5.46	2.76216	131375	9.10	2.79062	0.100
3	3.500	8.0	0.12500	3.34062	0.7660	6.13	3.38850	1.2000	9.60	3.41562	0.100
3 1/2	4.000	8.0	0.12500	3.83750	0.8210	6.57	3.88881	1.2500	10.00	3.91562	0.100
4	4.500	8.0	0.12500	4.33438	0.8440	6.75	4.38712	1.3000	10.40	4.41562	0.100
5	5.563	8.0	0.12500	5.39073	0.9370	7.50	5.44929	1.4063	11.25	5.47862	0.100
6	6.625	8.0	0.12500	6.44609	0.9580	7.66	6.50597	1.5125	12.10	6.54062	0.100

Nominal Pipe Size (inches)	Length,L1 Plane to L2 Plane External Thread (L2-L1)		Wrench Makeup Length for Internal Thread ⁷			Vanish Thread (V)		Overall8 Length External Thread (L4)	Nominal Complete External Thread ⁵		Height of Thread(h)	Increase in Diam./ Thread (0.0625/n)	Basic ⁶ Minor Diam.At Small End of Pipe(K0)
			Length(L3)		Diam. (E3)								
	Inch	Thread	Inch	Thread		Inch	Thread		Length. (L5)	Diam. (E5)			
1/8	0.1024	2.76	0.1111	3	0.35656	0.1285	3.47	0.3924	0.1898	0.37537	0.02963	0.00231	0.3339
1/4	0.1740	3.13	0.1667	3	0.46697	0.1928	3.47	0.5946	0.2907	0.49556	0.04444	0.00347	0.4329
3/8	0.1678	3.02	0.1667	3	0.60160	0.1928	3.47	0.6006	0.2967	0.63056	0.04444	0.00347	0.5676
1/2	0.2137	2.99	0.2143	3	0.74504	0.2478	3.47	0.7815	0.3909	0.78286	0.05714	0.00446	0.7013
3/4	0.2067	2.89	0.2143	3	0.95429	0.2478	3.47	0.7935	0.4029	0.99286	0.05714	0.00446	0.9105
1	0.2828	3.25	0.2609	3	1.19733	0.3017	3.47	0.9845	0.5089	1.24543	0.06957	0.00543	1.1441
1 1/4	0.2868	3.30	0.2609	3	1.54083	0.3017	3.47	1.0085	0.5329	1.59043	0.06957	0.00543	1.4876
1 1/2	0.3035	3.49	0.2609	3	1.77978	0.3017	3.47	1.0252	0.5496	1.83043	0.06957	0.00543	1.7265
2	0.3205	3.69	0.2609	3	2.25272	0.3017	3.47	1.0582	0.5826	2.30543	0.06957	0.00543	2.1995
2 1/2	0.4555	3.64	0.2500	2	2.70391	0.4337	3.47	1.5712	0.8875	2.77500	0.10000	0.00781	2.6195
3	0.4340	3.47	0.2500	2	3.32500	0.4337	3.47	1.6337	0.9500	3.40000	0.10000	0.00781	3.2406
3 1/2	0.4290	3.43	0.2500	2	3.82188	0.4337	3.47	1.6837	1.0000	3.90000	0.10000	0.00781	3.7375
4	0.4560	3.65	0.2500	2	4.31875	0.4337	3.47	1.7337	1.0500	4.40000	0.10000	0.00781	4.2344
5	0.4693	3.75	0.2500	2	5.37511	0.4337	3.47	1.8400	1.1563	5.46300	0.10000	0.00781	5.2907
6	0.5545	4.44	0.2500	2	6.43047	0.4337	3.47	1.9462	1.2625	6.52500	0.10000	0.00781	6.3461

Note: This information is selected from the International Standard for Pipe Threads, ASME B1.20.1.